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Future doctors' silence: An empirical study of declining speaking-up behavior among medical students

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ABSTRACT

Introduction: Speaking-up behavior is crucial for preventing patient harm, yet evidence suggests that increased clinical experience does not necessarily promote it. Gender, contextual factors, and prior patient safety training may also influence speaking-up tendencies, but research on their interaction remains limited. To address this gap, this study examines the impact of clinical experience, gender, patient safety training, and scenario severity on speaking-up behavior to inform targeted educational interventions.

Methods: This cross-sectional study analyzed 306 medical students from two cohorts (2020 and 2023) at a medical university in Northern Taiwan. A simulation-based assessment session evaluated students' speaking-up behavior in life-threatening and non-life-threatening medical error scenarios. Data on clinical experience, gender, and prior patient safety training were collected. Speaking-up behavior was assessed using the Speaking-Up Behavior Checklist, and binary logistic regression was conducted to examine associations.

Results: Increased clinical experience was significantly associated with lower speaking-up behavior (Exp(B) = 0.916, $p = 0.005$), particularly among male students, while female students' behavior remained unchanged. Additionally, students were less likely to speak up in life-threatening scenarios (Exp(B) = 0.886, $p = 0.007$), while scenario type had no significant effect on those with less clinical experience. Prior patient safety training was not significantly associated with speaking-up behavior.

Discussion: These findings suggest that clinical experience, gender, and scenario influence speaking-up behavior. While clinical skills improve with experience, heightened hierarchical awareness and professional relationships may increase hesitation. This implies the importance of structured policies, targeted training, and cultural transformation to foster a safe environment for speaking up. Future research should also examine psychological and organizational factors to better guide educational and systemic interventions.

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Introduction

Timely speaking-up behavior is crucial for preventing patient harm during medical errors [1,2]. As junior healthcare providers gain clinical experience, they typically develop greater management and communication skills [3,4], which, in theory, should enable more effective speaking up [5,6]. However, studies from the past decade report wide variability in speaking-up rates, ranging from 20% to 83% [1,7–13]. The variation in speaking-up rates in response to medical errors reflects the complex interplay of individual, team, organizational, and cultural influences [14,15]. Several models have been used to examine the factors influencing speaking-up behaviors in response to medical errors across individual [10,16], team [17–19], organizational [20–22], and cultural levels [23]. Across cultural, organizational, team, and individual levels, these models indicate that supportive climates, inclusive leadership, and favorable work environments foster psychological safety and situation monitoring, mitigate discrimination and silence, promote voice and error reporting, and ultimately enhance patient safety and quality of care. However, many studies do not fully account for individual characteristics of medical students, which play a critical role. Strengthening these models requires greater emphasis on key predisposing factors such as gender [7,24], age [7], patient

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Practice points

- Male medical students showed a significant decline in speaking-up behavior as their clinical experience increased.
- Among male students, the likelihood of speaking up was lower in life-threatening scenarios compared to non-life-threatening ones.
- These patterns were not observed in female students, whose speaking-up behavior remained stable regardless of clinical experience or scenario severity.
- These findings call for structured interventions—clear policies, targeted training, and cultural change—to encourage speaking up and protect patient safety.

safety training [24], educational background [9], clinical experience [10,25], and clinical role [9,13,25], as these factors may fundamentally influence or interact with other contextual and psychometric factors. Moreover, most empirical evidence on the relationship between individual factors and speaking-up behaviors relies on questionnaire-based surveys or qualitative approaches [7,9,10,13–15,24,25], highlighting that objectively measuring speaking-up behaviors remains challenging.

Exploring the relationship between these predisposing factors and medical students' speaking-up behaviors is important for educational purposes, given its relevance to patient safety. For instance, gender-related behaviors, shaped by observers' expectations, self-concept, and situational cues, may impact students' willingness to speak up yet research on gender as a determinant of speaking-up behavior remains limited [26]. Time-related factors such as age and clinical experience are often correlated with each other, and with other patient safety and teamwork factors [27,28] but, again, research into the association between these factors and speaking up is limited. Patient safety training enhances students' knowledge and attitudes toward safer care [29,30] but it's effect on actual speaking-up behavior remains unclear. Finally, while several systematic reviews have examined situational factors influencing speaking up [31,32], little is known about how the severity of a patient's condition may affect students' likelihood of intervention. Given the importance of speaking up in patient safety, further research is needed to address these gaps [33].

While the authors' previous study primarily evaluated the effectiveness of personalized speaking-up training, it also noted gender differences in responses to medical errors [34]. Building on this, the present study focuses on baseline characteristics that may shape students' speaking-up behaviors in medical error scenarios prior to training, with the aim of providing insights for medical educators on integrating assertive communication training at optimal stages and developing targeted educational materials that account for gender differences and scenario variations. Our hypotheses were that (a) gender, (b) age, (c) clinical experience, (d) patient safety training, and (e) severity of medical error scenario can be associated with speaking-up behaviors.

Method

This simulation-based cross-sectional study was based on data from a routine training course with structured and standardized scenarios at a medical university in Northern Taiwan. The training course consisted of multiple components, including pre-course material review, background knowledge discussions and assessments, a pre-intervention simulation scenario, a personalized debriefing session, and a post-intervention simulation scenario practice. The research team obtained approval from the Institutional Review Board to conduct the study within the course.

Settings and participants

Data for the present study were collected in 2020 and 2023. The training course was suspended in 2021 and 2022 due to COVID-19 prevention measures. Part of the pre-intervention evaluation data had also appeared in previous studies, as they were based on the same participants and settings [35]. The participating medical students were in their final two years of 6-year medical school curriculum and took part in the simulation activity during their clinical rotation training. Before engaging in the medical error simulation scenario, the study team ensured that students had acquired sufficient background knowledge of the clinical condition presented in the scenario and fully understood the principles of the simulation activity.

Demographic variables of participants

A set of demographic data was systematically gathered from medical students before the pre-intervention simulation scenario, including their gender, age, extent and type of clinical experience, and history of involvement in patient safety training programs. This information was compiled using a

carefully designed, structured questionnaire tailored to provide an in-depth understanding of each participant's personal and professional background.

Evaluation of speaking-up behavior in medical errors

Medical students' speaking up behaviors were evaluated during their involvement in two scenarios related to medical errors, which included the use of a full-body computerized mannequin, a standardized nurse, and a standardized senior physician. The scenarios and the operational protocol have been validated as an effective approach for evaluating medical students' speaking-up behaviors in medical errors [34,35]. The scenario was designed to either present two distinct clinical situations: one with stable vital signs, representing a nonlife-threatening condition, and another with unstable vital signs, indicating a life-threatening situation. Although the scenario involved non-life-threatening conditions for some medical students and a life-threatening situation for others, all students were initially instructed to (a) assess the patient's medical condition by interacting with the standardized nurse and the computerized mannequin at the start of the simulation, and (b) prescribe the appropriate medication, which the standardized nurse then prepared after making an assessment. As the standardized nurse was preparing the medication, the senior physician entered the simulation. Following a brief exchange with the nurse and the student, the senior physician mistakenly ordered the wrong medication for the patient. If the student remained silent about the medical error, the nurse proceeded to administer the incorrect medication. However, if the student spoke up and expressed concern about the error, the senior physician engaged in a discussion with the student regarding their treatment recommendation, in accordance with standardized scripted responses. To standardize the study, the standardized nurse and senior physician were both assigned female genders.

Assessment instruments, pilot test, and rater calibration

Speaking-up behavior in medical errors was evaluated using the Speaking-up Behavior Checklist in simulation scenarios [34]. This checklist is a modified version of the Pian-Smith Evaluation Tool [36], and it had been previously adapted and demonstrated construct validity in related studies [34,35]. The tool uses a 5-point rating scale: 1 indicates no reaction, 2 suggests negative facial or body language, 3 involves making an indirect or vague statement, 4 entails advocating or asking questions, and 5 represents persistent advocacy or inquiry, initiating discussion. An expert panel comprising one simulation expert, two emergency attending physician, one nurse practitioner, and one registered nurse reviewed the adaption and agreed that a score of 4 or higher should be considered clear speaking-up behavior in response to medical errors. Speaking up was considered successful if it occurred before the standardized nurse administered the incorrect medication to the mannequin, as prescribed by the standardized senior physician.

For the present study, pilot simulation scenarios were conducted and recorded to refine the adaptation and to facilitate rater training. Two raters independently reviewed the pilot recordings, practiced applying the checklist to speaking-up behaviors, and engaged in consensus discussions to calibrate scoring and clarify coding rules. After this calibration, they independently assessed medical students' speaking-up behavior in real-time simulation scenarios. Interrater reliability, measured using Krippendorff's alpha, was 0.851 in 2020 and 0.923 in 2023, indicating a high level of agreement [37].

Statistical analysis

Data analysis was mainly conducted using SPSS version 20.0 (IBM Corp., Armonk, NY, USA). Descriptive statistics were used to summarize demographic and study variables, with frequencies and percentages reported for categorical variables and means with standard deviations for continuous variables. To examine relationships between students' speaking-up behavior, and gender, age, clinical experience and prior patient safety training. Spearman's rho correlation coefficient was applied as data violated normality assumption. Additionally, logistic regression analysis including both univariate and multivariate models was performed to assess the association between speaking-up behavior and the actual predisposing factors. Since gender and the severity of medical error situations can affect speaking-up behaviors as demonstrated in our previous studies [24,34], the data underwent additional analysis based on the different strata of gender and severity. We checked residuals of each predictor and fitted values and rigorously identified outliers in the logistic regression analysis using Cook's distance [38], studentized residuals, and hat-values. Due to the outliers and poor fitness, we further applied a Bayesian generalized linear model with Cauchy prior to check whether the results were robust [39]. All statistical analyses were conducted with a 95% confidence interval, and statistical significance was set at $p < 0.05$.

Results

A total of 306 medical students who completed the simulation course were included in this study,

comprising 146 students from the 2020 cohort and 160 students from the 2023 cohort. Participant data were missing for one student's age, one student's gender, and five students' prior patient safety training, and these cases were excluded from the analysis. Among those with available data, the gender distribution was 186 males (61.0%) and 119 females (39.0%). The participants had a mean age of 23.88 ± 1.47 years, ranging from 22 to 33 years. Their clinical experience averaged 13.33 ± 4.06 months, with a range of 6 to 19 months. Regarding the simulation scenarios, 148 students (48.4%) participated in a life-threatening scenario, while 158 students (51.6%) experienced a non-life-threatening scenario. Additionally, 238 students (79.1%) reported previously receiving patient safety training. Detailed demographic information is presented in Table 1. A correlation analysis using Spearman's rho revealed a moderate positive correlation between age and clinical experience ($\rho = 0.396$, $p < 0.001$), suggesting that these two variables are interrelated (Table A1).

The findings of univariate logistic regression showed that age and clinical experience were significant predictors of speaking-up behavior. Specifically, greater age was negatively associated with speaking-up behavior (Exp (B) = 0.805, 95% CI: 0.66 to 0.99, $p = 0.04$), meaning that older students were less likely to speak up (Table 2). Clinical experience was also negatively associated with speaking-up behavior (Exp (B) = 0.922, 95% CI: 0.87 to 0.98, $p = 0.007$), indicating that students with greater clinical experience exhibited lower likelihoods of speaking up. Gender, previous patient safety training, and severity of medical error scenario were not significantly associated with speaking-up behavior ($p = 0.61$, $p = 0.88$, and $p = 0.31$, respectively).

The findings of multivariate logistic regression showed that clinical experience was a significant negative predictor of speaking-up behavior (Exp (B) = 0.929, 95% CI: 0.87 to 0.99, $p = 0.02$), while age was not significantly associated with speaking-up behavior (Exp (B) = 0.874, 95% CI: 0.71 to 1.07, $p = 0.20$) (Table 3). A condition index of 53.37 indicated multicollinearity with concern with age contributing most significantly to this concern (a variance proportion of 0.99 in the dimension corresponding to CI = 53.37). To better isolate the independent effect of clinical experience on speaking-up behavior, an additional regression model was run excluding age. In this adjusted model, clinical experience remained a significant negative predictor of speaking-up behavior (Exp (B) = 0.916, 95% CI: 0.86 to 0.97, $p = 0.005$), with higher clinical experience being associated with a lower likelihood of speaking up.

Other variables, including gender, previous participation in patient safety training, and scenario type, did not show statistically significant associations with speaking-up behavior in either model. The results of the multivariate analysis, including interaction effects between speaking-up behavior and clinical experience, are presented in Appendix Table A2. Although initial analyses suggested concerns about residual fit and outliers (Appendix Figures 1 and 2), robust analyses confirmed that the statistical significance and overall direction of effects remained consistent (Appendix Figure 3 and Appendix Table A3).

Considering the real differences in gender and the severity of medical error scenarios, additional analyses were carried out for each distinct group (Table 4). The findings from multivariate logistic regression analyses indicated that clinical experience was a significant negative predictor of speaking-up behavior among male participants (Exp (B) = 0.875, 95% CI: 0.81 to 0.95, $p = 0.001$) and in life-threatening scenarios (Exp (B) = 0.886, 95% CI: 0.81 to 0.97, $p = 0.007$). In contrast, none of the examined factors showed a statistically significant association with speaking-up behavior among female participants and in non-life-threatening scenario.

Table 1. Participant demographics ($N = 306$).

Characteristics	*N*	Cohort
Age (year), Mean ± SD (range)	305	23.88 ± 1.47 (22–33)
Male/Female n (%)	305	186 (61.0)/119 (39.0)
Clinical experience, Mean ± SD (month)	306	13.33 ± 4.06 (6–19)
Experience scenario (life-threatening scenario/non-life-threatening scenario)	306	148 (48.4)/158 (51.6)
Received training course related to patient safety (Yes/No), n (%)	301	238 (79.1)/63 (20.9)

Table 2. Univariate analysis of potential factors on speaking-up behavior ($n = 306$).

			95% CI		
Variables	N	Exp (B)	Lower	Upper	*p*
Gender	305	1.135	0.69	1.86	0.61
Age	305	0.805	0.66	0.99	0.04*
Clinical Experience	306	0.922	0.87	0.98	0.007*
Received training course related to patient safety	301	1.048	0.58	1.90	0.88
Scenario	306	1.279	0.79	2.06	0.31

Note: Dependent variables: speaking-up behavior. Enter (Regression). *$p < 0.05$. Two tailed.

Table 3. Multivariate analysis of potential factors of speaking-up behavior ($n = 301$).

	All variables				Without age			
		95% CI				95% CI		
Variables	Exp (B)	Lower	Upper	*p*	Exp (B)	Lower	Upper	*P*
Gender	1.184	0.71	1.97	0.52	1.140	0.69	1.89	0.61
Age	0.874	0.71	1.07	0.20	–	–	–	–
Clinical Experience	0.929	0.87	0.99	0.02*	0.916	0.86	0.97	0.005*
Received training course related to patient safety	1.086	0.59	2.00	0.79	1.145	0.63	2.10	0.66
Scenario	1.134	0.69	1.86	0.62	1.179	0.72	1.93	0.51

Note: Dependent variables: speaking-up behavior. Enter (Regression). Missing data: 5. All variables (Cox & Snell R^2: 0.035, Nagelkerke R^2: 0.049). Without age (Cox & Snell R^2: 0.030, Nagelkerke R^2: 0.041). $^*p < 0.05$. Two tailed.

Table 4. Multivariate analysis of potential factors of speaking-up behavior by gender and severity of scenario.

		95% CI		
Variables	Exp (B)	Lower	Upper	*p*
Male				
Clinical Experience	0.875	0.81	0.95	0.001*
Received training course related to patient safety	1.425	0.67	3.05	0.36
Scenario	1.325	0.70	2.51	0.39
Female				
Clinical Experience	0.971	0.89	1.06	0.53
Received training course related to patient safety	0.806	0.29	2.24	0.68
Scenario	1.052	0.47	2.33	0.90
Life-threatening scenario				
Gender	1.301	0.61	2.78	0.50
Clinical Experience	0.886	0.81	0.97	0.007*
Received training course related to patient safety	1.037	0.45	2.40	0.93
Non-life-threatening scenario				
Gender	1.008	0.50	2.02	0.98
Clinical Experience	0.946	0.87	1.03	0.20
Received training course related to patient safety	1.323	0.54	3.24	0.54

Note: Dependent variables: speaking-up behavior. Enter (Regression). $^*p < 0.05$. Two tailed.

Discussion

This study identifies clinical experience as a key factor influencing medical students' speaking-up behavior in response to medical errors. The overall findings suggest that as students gain clinical experience, their likelihood of speaking up decreases. However, the impact of clinical experience on speaking-up behavior varied between genders. Among male students, greater clinical experience was associated with a significant reduction in speaking up, whereas female students showed no significant change. Further analysis of simulation scenarios revealed that in life-threatening situations, increased clinical experience corresponded to a lower likelihood of speaking up, while in non-life-threatening scenarios, clinical experience had no significant effect.

Previous studies indicate that senior trainees are more adept at identifying violations or potential patient safety issues than their junior counterparts [13,40], suggesting that accumulating clinical experience enhances sensitivity to patient safety concerns. However, despite this increased awareness, speaking-up behavior does not necessarily improve with experience. A survey-based study reported a significant increase in students choosing to remain silent as their training progressed [13]. The present study further demonstrates that in standardized, controlled scenarios, students with greater clinical experience were less likely to speak up when confronted with medical errors. Studies highlight the complexity of speaking-up decisions [24,33]. Researchers found that "emotional maturity" influenced how trainees addressed potential errors [40]. Rather than directly challenging mistakes, they often expressed concerns indirectly or remained silent to navigate professional relationships and hierarchical pressures. Researchers also found that upper-year students reported lower confidence in teamwork and safety culture competencies compared to their junior peers [41], suggesting that increased clinical exposure heightens awareness of hierarchical structures, organizational constraints, and the risks associated with speaking up [14,42]. In the Taiwanese context, hierarchy, deference to authority, and face-saving are pervasive cultural features that may further discourage students from speaking up as they gain seniority. Power distance, defined as the extent to which

unequal power distributions are expected and accepted [43], is particularly salient in East Asian societies and has been linked to reluctance to question superiors in healthcare settings [44,45]. Recent evidence highlights power distance as a multilevel phenomenon influencing communication, psychological safety, and error reporting in healthcare [45], providing an additional cultural lens through which the present findings can be interpreted. This highlights the importance of integrating assertiveness and communication training into medical education to ensure that increased clinical experience does not diminish advocacy for patient safety [46].

The current study found that male medical students became significantly less likely to speak up as their internship experience increased, a pattern not observed among female students. Previous research on gender differences in speaking-up behavior has yielded mixed results. Some studies report higher speaking-up rates among male students and clinical trainees [7,23], attributing this to greater confidence and fewer internal barriers. Other studies have reported the opposite. For example, one previous study found that male trainees were significantly less likely to speak up [16]. Most prior studies were based on self-reported data, whereas the present study is based on direct behavioral observation in simulated clinical scenarios, offering a different perspective on how gender and clinical experience influence speaking-up behavior. This trend may reflect differences in decision-making styles. Research suggests that medical students often display styles such as hypervigilance and buck-passing [47], and that male students are more likely to adopt a procrastination style. These tendencies may contribute to hesitation in hierarchical or uncertain clinical situations. As clinical experience accumulates, increasing awareness of professional dynamics and perceived interpersonal risks may further discourage speaking up among male students. These findings point to the need for further research into gender-related influences on speaking-up behavior. A clearer understanding of these patterns can help inform the development of targeted educational and policy interventions to support open communication and enhance patient safety.

Contextual factors influence speaking-up behavior. A survey found that 40% of students remained silent when witnessing high-risk patient situations [7]. The present study further indicated that as clinical experience increases, the likelihood of speaking up in life-threatening situations declines, which differs from findings of a previous survey-based study [16]. These discrepancies may reflect distinctions in methodology, as self-reported surveys may not capture actual behavior under pressure. Learner roles and backgrounds may also be relevant. Medical students tend to defer to authority, whereas experienced professionals may leverage their expertise to navigate high-stakes situations more effectively [48,49]. These factors likely contribute to variation in findings. While this study contributes to the literature by examining speaking-up behavior, gender, and clinical experience, further research is needed to validate these findings in real-world settings, where hierarchical structures and professional dynamics further shape students' willingness to speak up.

Our previous study found that male medical students were more likely than female students to speak up in life-threatening medical error scenarios [34], whereas the current study showed different patterns. These discrepancies may reflect differences in learner stage, curriculum system, and the impact of COVID-19. The earlier study among both sixth-year clerks and seventh-year interns in Taiwan's 7-year curriculum system conducted in 2018 before the pandemic. In contrast, the current study included fifth- and sixth-year clerks in the newer 6-year curriculum system during the early pandemic (2020) and the post-pandemic phase (2023). Prior work has shown that students in the 7-year curriculum system demonstrated stronger clinical skills and higher OSCE scores than those in the 6-year curriculum system [50,51]. Pandemic-related reductions in clinical exposure and reliance with online learning may also have affected competency development [52,53]. Taken together, variations in clinical training experience, study aim, curriculum, and learning context may help explain the divergent findings on speaking-up behavior across studies. How these mechanisms intersect with gender and training stage remains unclear, and further research is needed to clarify these dynamics.

This study also highlights not only the significant roles of clinical experience in medical students' likelihood of speaking up in response to medical errors, but also the heterogeneous patterns of the association by gender and severity of scenario. The findings imply that the influence of actual clinical context on speaking-up behavior may outweigh the impact of patient safety training courses. This suggests the need to redesign these courses to more effectively promote behaviors related to patient safety, particularly speaking up on behalf of patients. For instance, by designing tailored learning experiences that accommodate diverse trainee profiles, educators can improve learning outcomes, strengthen the culture of patient safety, and equip medical students with the confidence and skills needed to speak up when it matters most [34].

Strengths and limitations

Through standardized and structured observation, our study strengthens the evidence on this topic, as most previous studies have relied on self-reported surveys. Additionally, the present multivariate analysis enhances understanding of the association between actual predisposing factors and medical students' speaking-up behaviors in response to medical errors. This study's generalizability is limited by its single-site design, conducted at a medical university in Northern Taiwan, which may affect its applicability to other cultural and educational settings. Additionally, the analysis was restricted to two cohorts (2020 and 2023), potentially overlooking variations in speaking-up behavior across a broader student population. Further details on the type, duration, and timing of patient safety training are needed to better evaluate its impact on speaking-up behavior. While simulation-based assessments offer a standardized approach, they may not fully capture students' real-world responses in clinical settings. Additional evidence of construct validity for the assessment tool was not carried out beyond that reported in prior work. Moreover, this study did not account for individual psychological factors, such as personality, risk perception, and prior speaking-up experiences, which may influence decision-making in medical error situations. Cross-cultural and comparative research is warranted to better understand contextual variations in speaking-up behaviors.

Conclusions

This study suggests that increased clinical experience may reduce speaking-up behavior, particularly among male students and in life-threatening situations, emphasizing the need for targeted interventions. Future research should explore the underlying mechanisms of these factors to develop personalized training strategies that strengthen patient safety advocacy in medical education.

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Author contributions statement

Hui-Wen Chen, Yi-No Kang, and Jen-Chieh Wu: conceptualization. Hui-Wen Chen, Yi-No Kang: formal analysis. Yi-Chun Chen and Hung-Wei Tsai: investigation. Hui-Wen Chen, Yi-No Kang, Jen-Chieh Wu: methodology. S. Barry Issenberg and Jen-Chieh Wu: supervision. Hui-Wen Chen and Jen-Chieh Wu: writing– original draft. Hui-Wen Chen, Yi-Chun Chen, Hung-Wei Tsai, Enoch Yi-No Kang, Jen-Chieh Wu, and S. Barry Issenberg: writing– review & editing.

Disclosure statement

Hui-Wen Chen, Yi-Chun Chen, Hung-Wei Tsai, Enoch Yi-No Kang, Jen-Chieh Wu, and S. Barry Issenberg states that they have no financial or non-financial conflicts of interest to disclose related to this manuscript.

Declaration of use of generative AI

During the preparation of this manuscript, the authors utilized ChatGPT-4o to assist with improving readability and language clarity. No content was generated by the AI in place of the authors' original ideas or analysis. All AI-assisted text was thoroughly reviewed and edited by the authors prior to submission. The authors accept full responsibility for the final content of the manuscript.

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Data availability statement

The data generated and/or analyzed in this study can be obtained from the corresponding author upon reasonable request.

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Appendix

Table A1. Correlation coefficients (Spearman's rho) among age, and clinical experience.

Variables	Age	Clinical experience
Age	1 ($N = 305$)	
Clinical Experience	0.396 $p < 0.001$* ($N = 305$)	1 ($N = 306$)

Note: *$p < 0.05$. Two tailed.

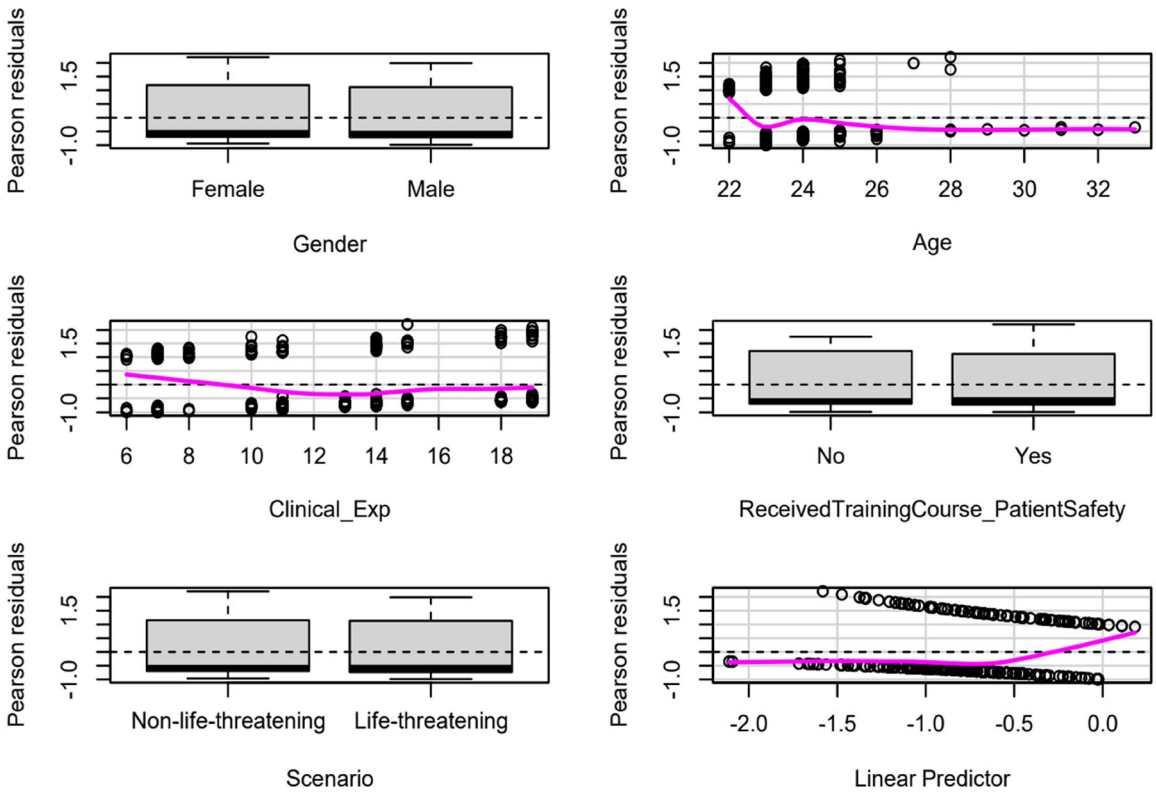
Table A2. Multivariate analysis of factors influencing speaking-up behavior.

		95% CI		
Variables	Exp (B)	Lower	Upper	*p*
Gender	4.039	0.75	21.84	0.11
Clinical Experience	1.281	0.87	1.89	0.21
Received training course related to patient safety	1.120	0.61	2.07	0.72
Scenario	2.298	0.43	12.21	0.33
Age x Clinical Experience	0.990	0.98	1.01	0.19
Gender x Clinical Experience	0.909	0.80	1.03	0.13
Scenario x Clinical Experience	0.947	0.84	1.07	0.39

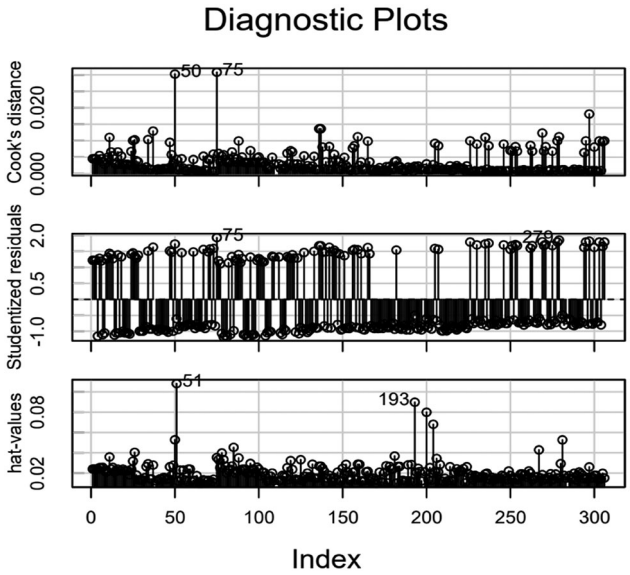
Note: Dependent variables: speaking-up behavior. Enter (Regression). Missing data: 5. Cox & Snell R^2: 0.047. Nagelkerke R^2: 0.067. *$p < 0.05$. Two tailed.

Table A3. Model estimates for predictors of speaking-up behavior.

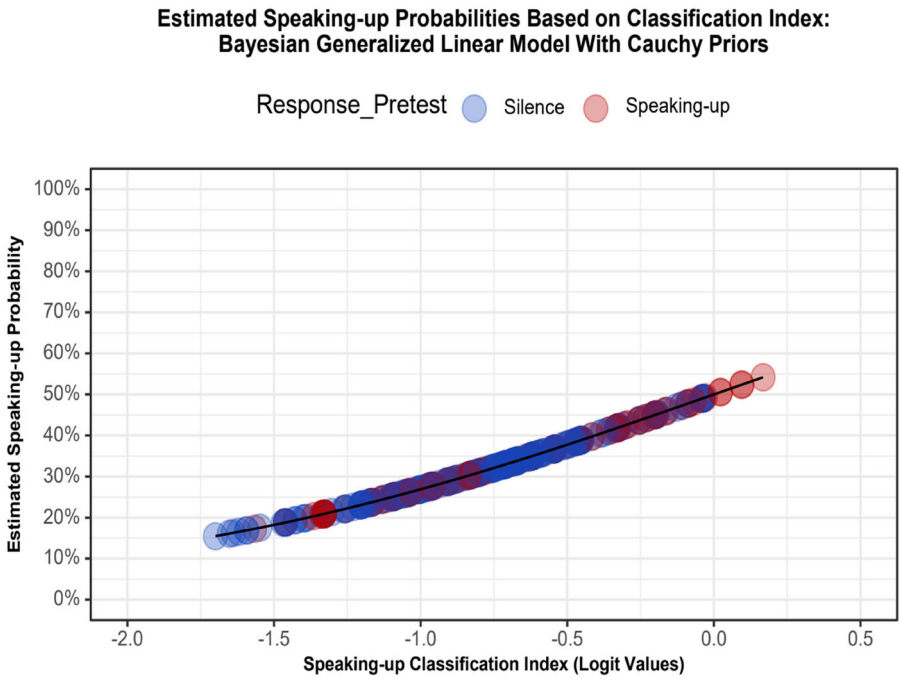
Item	Estimate	Std. Error	z value	Pr (>|z|)
INTERCEPT	3.139	2.384	1.316	1.88e-01
Gender (Male)	0.165	0.258	0.639	5.23e-01
Age	−0.132	0.102	−1.291	1.97e-01
Clinical experience	−0.073	0.032	−2.258	2.40e-02
Received training course (Patient safety)/Yes	0.079	0.306	0.259	7.96e-01
Scenario (Life-threatening)	0.124	0.25	0.497	6.19e-01



Figures A1. Diagnostic plots of Pearson residuals across predictors for assessing model fit.



Figures A2. Diagnostic plots showing Cook's distance, studentized residuals, and hat-values for evaluating potential outliers, leverage, and influential observations in the regression model.



Figures A3. Estimated probabilities of speaking-up responses from a Bayesian generalized linear model.